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DESIGN ISSUES FOR THE POWER SYSTEM OF A LUNAR ROVER

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ABSTRACT

The Electrical Power System developed for a lunar rover is presented and the design of the main subsystems is analysed with a view to the AMALIA lunar mission. In particular, an unusual configuration of the battery subsystem, based on a 1S battery and a high-voltage-diversity BCDR was developed and tested. The experimental results demonstrated that any bus voltage in the 12-48V range can be managed with a satisfactory conversion efficiency by using a non-insulated TLNPC converter. Other design issues of the main EPS units are discussed.

1. INTRODUCTION

This work deals with the study of the power system of a rover for lunar exploration, carried out within the activities related to the AMALIA lunar mission, aimed to participate to the Google Lunar X - prize.
Electrical power systems of rovers differ from those used in orbiting spacecrafts mainly because of the unpredictable power profile, depending on the surface details. Actually the absorbed power significantly varies along with the rover's motion and battery intervention may be required to support primary source during load peaks created by slopes and rocks. In a vehicle intended to operate on the Moon in a nearly equatorial environment, battery may be also required to supply full power for a short time, just to allow the rover to pass through a shaded area.
The PCDU in use was derived from that previously developed for a satellite which was launched in 2012 by VEGA0 and operated for some days, transmitting telemetry and a beacon radio signal. The lesson learned by this short mission suggested some modifications intended to increase reliability of the modules internal to the PCDU. Among these the qualification level of most components was raised from COTS to standard military, while no rad-hard component was introduced, due the short duration scheduled for the mission. Actually the most challenging environmental constraints derive from thermal issues.
The new design also incorporates a new BCDR module, based on a high voltage gain bidirectional non-insulated converter: it demonstrates the possibility to interface a standard bus with a single-cell Li-Ion battery (more precisely a 1S configuration), simplifying the battery harness and eliminating any problem originated from mismatching of series-connected cells [1] [2]. Such single cell battery only can be used if: 1) the handled power is small enough to avoid current to exceed feasibility limits, 2) a converter with a suitable voltage ratio is available. Small rovers, with power in the tens of Watt range, satisfy requirement 1). The proposed converter satisfies requirement 2). Therefore, a series of demonstrators have been designed and tested to verify the feasibility of the proposed approach.
The paper, after a brief outline of the general configuration of the AMALIA rover, examines the main issues related to the design of the single units forming the EPS system.

2. SYSTEM CONFIGURATION AND DESIGN DRIVERS

A preliminary configuration of the AMALIA rover is reported in Fig. 1 [3]. A four-wheels locomotion system provides the capability to explore a planetary surface, under the control of an optical guidance system. In order to enhance both the visual and the communication ranges the cameras and the antennas are both placed on top of a mast.
The Electrical Power System (EPS) includes a Solar Array (SA) as a primary energy source, a Li-Ion battery as a secondary energy source, a Power Converting Unit

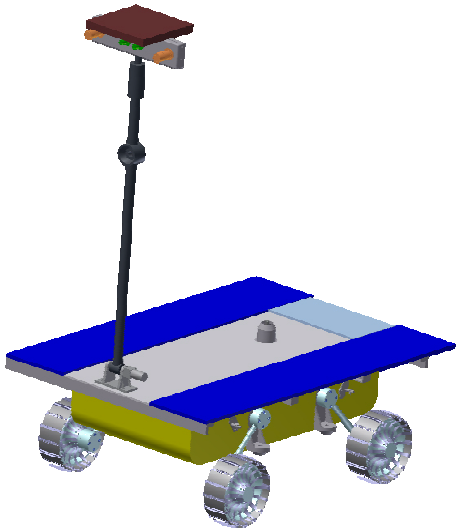


Figure 1. Preliminary configuration of the Amalia Rover

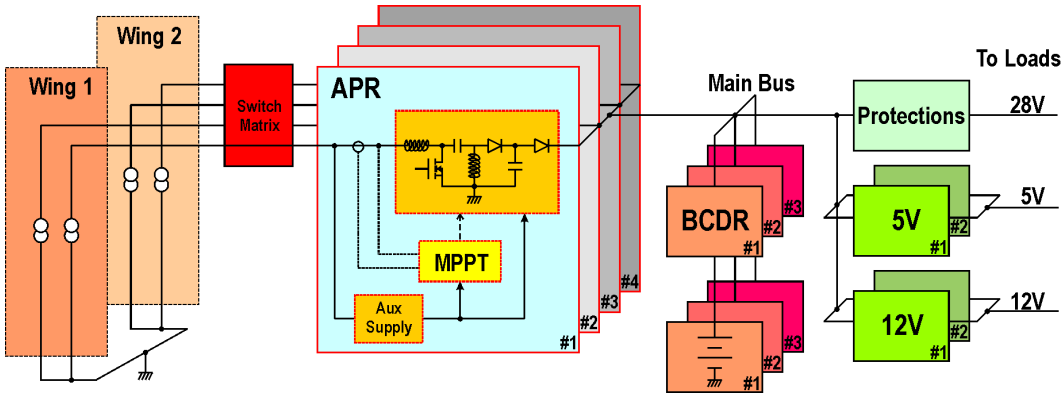


Figure 2. A schematic of the proposed PCDU for the AMALIA lunar rover

(PCU) comprising 4 autonomous Array Power Regulators (APR) and a Power Distribution Unit (PDU) generating payload buses at 28V, 12V and 5V (Fig. 2).
It is designed to ensure a minimum daily operation time corresponding to the visibility of a single Earth Station for at least 8 consecutive terrestrial days and no device for surviving the lunar night was included [4].

2.1 Solar Array

The SA is split in four sections, each consisting of a string of triple junction solar cells. In nominal conditions each section supplies a dedicated APR, capable to perform an independent Maximum Power Point Tracking (MPPT). This arrangement was selected for its robustness with respect to string unbalances due to cell shading (quite likely with low un elevation) or failures. In fact, the event of missing the whole power generation from a string would be unacceptable as the SA was sized to satisfy peak power absorption under irradiation with 90° Sun elevation and the only margin for graceful performance degradation is provided by the gap between peak and stationary power demand. On the other hand, at the maximum load absorption the battery S/S is always expected to complement the SA power. Therefore a partial failure of the SA could reduce the operational time and extend the battery recharging time. The relations among operation time, battery capacity and available power from SA are illustrated by the timelines reported in Fig. 3:

2.2 Power Converting Unit

The SA is interfaced to the Main Bus by a PCU comprising 4 APRs and a switch matrix.
As the APR circuits were designed to manage a power double than the maximum expected from a single SA section, this approach provides a redundancy larger than 2/4. The problem of avoiding SPF in the switch matrix was analysed in [5].
The APRs are based on SEPIC converters, whose operation is managed by independent MPPT controllers (Fig. 4), based on the patented SMP technique [6]. This approach was preferred to the well known Sequential Switching Shunt Regulation (S3R), mainly because it is capable to manage more effectively the frequent solar cell shading by the mast. Actually, in a S3R system the cell shading would be faced only by introducing in any string a few additional solar cells with respect to the minimum required to generate the nominal voltage.

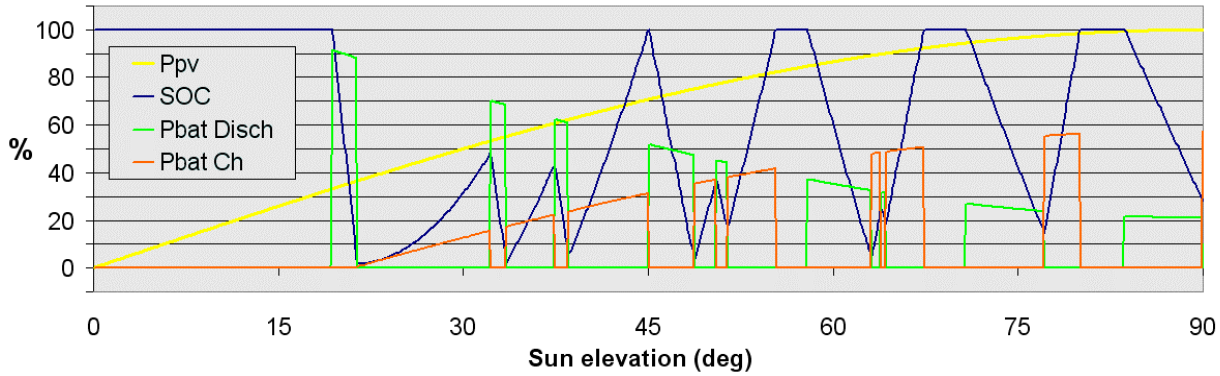


Figure 3. Power timeline for SA and battery.

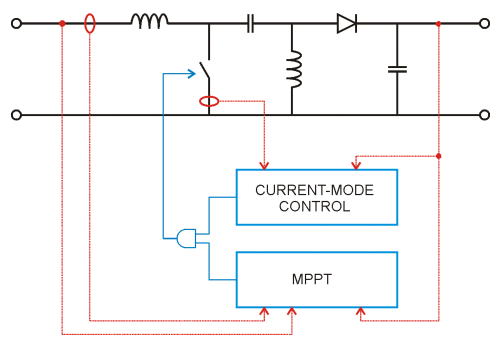


Figure 4. SEPIC converter with current-mode control

The switch matrix was included in order to increase the reliability against APR failures [5]. Actually, it allows to connect all SA sections at the input of any APR. This condition is sketched in Fig. 5, where the failure of an APR is managed by paralleling two SA sections.

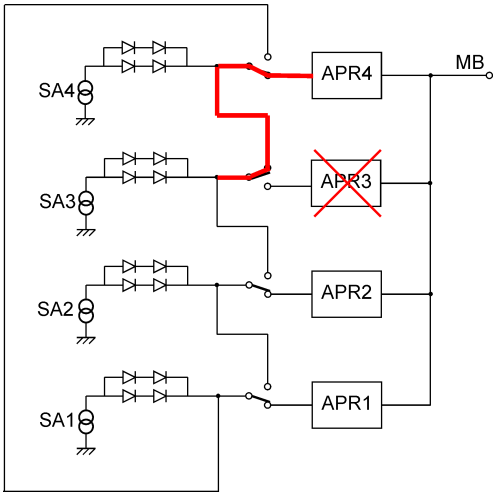


Figure 5. The switch matrix. A failure in APR3 is managed by switching the SA3 output to APR4.

As a further advantage, the MPPT-based APR took advantage from the experience gained with a PCDU previously developed and flown [5].
The design upgrades with respect to the previous PCDU board were based on a revision of the circuit design, more than on an analysis of the on-flight behaviour. In fact scarce telemetry data were downloaded only in one passage in sunlight and one in eclipse, giving no clear evidence of any trouble.
Though the qualification level of most components is set at the standard military grade, some COTS components are still kept, but their interface with the rest of the circuit was redesigned in order to increase the capability to contain the effects of possible failures. For example, a weak point was singled out in the COTS microcontroller for MPPT, which was prone to failure modes capable to stop the whole APR. Therefore the interface between the microcontroller and the converter was redesigned in order to enable the microcontroller to override the basic duty-cycle generation circuit, inspired to [7], only when it is correctly operating. The thermal design of the board was completely revised.

2.3 Bus and Power Distribution

The nominal main bus (MB) voltage was set to 28V. When excess power is available from the SA, the voltage is regulated by controlling the output current from the APRs. During battery charging, if the APRs are in MPPT, the control of the MB voltage passes to the feedback circuit outlined in Fig. 6, which modulates the battery current on the basis of the available excess power from the PV system. The experimental trend of the battery current with respect to the bus voltage is reported in Fig. 7.
As will be discussed subsequently, two designs were developed for the battery subsystem, providing sun-regulation or full regulation. In both cases the regulated payload buses at 12V and 5V are derived from the MB by means of a pair of redundant buck converters. All payload buses are protected against overcurrents by means of suitable Latching Current Limiters.

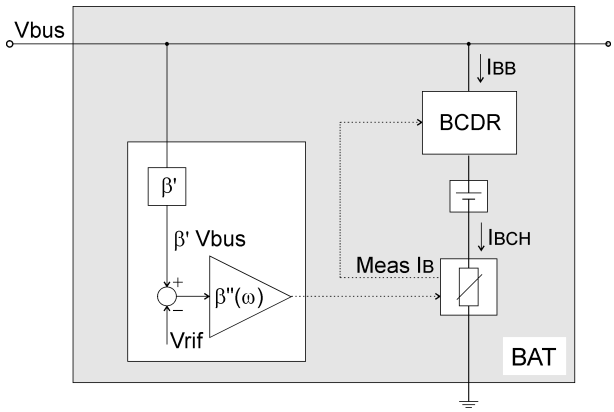


Figure 6. Block diagram of the control circuit for battery charging current

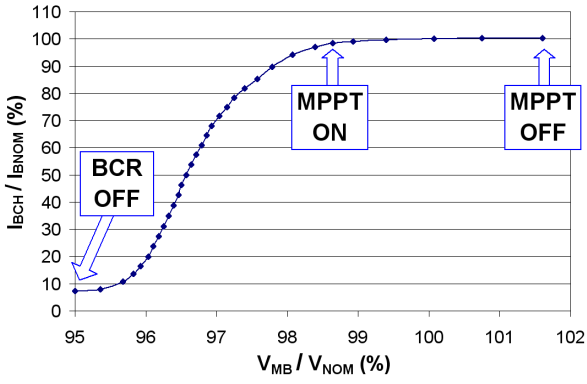


Figure 7. Charging current Vs. bus voltage

2.4 Battery subsystem

The battery subsystem is mainly designed to complement the primary power generator. It comprises three units in redundancy 2/3, each consisting of a battery and its BCDR. Two designs were developed for the battery units. A first design is based on a classical approach with a 8S Li-Ion battery (manufactured by ABSL) and a sun-regulated 28V bus.
The alternative design is based on a 1P1S configuration, consisting of a single VES cell (manufactured by SAFT), which is interfaced to the MB voltage by a high-gain bidirectional converter. This solution provides several benefits. 1) Flexibility, as the MB voltage can be selected in a wide range. 2) Drastic simplification of the internal battery harness. 3) Simple battery cell managing, as the approach is intrinsically unaffected by cell mismatching, so that no cell balancing device is required. (This benefit would be more appreciated in other applications, like for example LEO orbits, where the large number of cycles could increase stress on the mismatched battery cells and eventually lead to anticipated failure.) 4) Power losses are slightly larger than in a sun-regulated bus in the case of 28V MB voltage, but they are nearly halved if 12V is selected for the MB. 5) The regulated bus voltage simplifies the payload interfaces. 6) As the same bidirectional circuit can operate both as a BCR and as a BDR, the complexity increase is negligible.

2.4.1 The BCDR converter

A BCDR demonstrator was built and extensively tested in order to demonstrate the possibility to interface any bus voltage in the range 12-48V with a single Li-ion cell discharged down to 3 V, with up to 30A battery current [8].
The circuit is based on a non-insulated bidirectional converter with the Three Level Neutral Point Clamped (TLNPC) topology (Fig. 8), well known for its intrinsic capability to provide high voltage ratios and/or high efficiency [9].

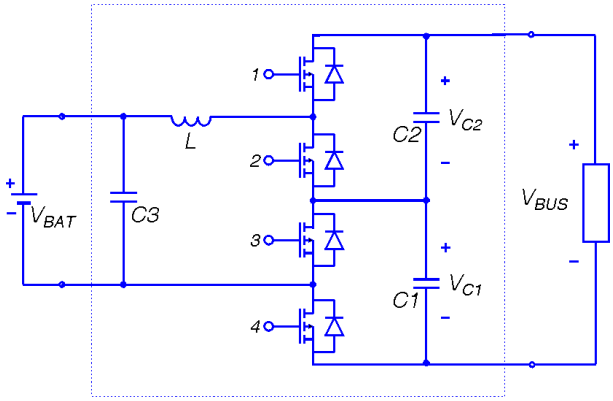


Figure 8. Schematic of the TLNPC non-insulated Bidirectional converter

The most challenging operation mode is in BDR, when the converter is required to boost the battery voltage, in the 3-4.2V range to the 28V main bus voltage. Actually, as shown in Fig. 9, a breadboard based on automotive components demonstrated 3/28V operation with an efficiency larger than 92% and up to 94% efficiency at higher battery voltages. The best efficiency was achieved in the 10-20A range. A 12V bus voltage is less favourable from an efficiency point of view, as similar results are achieved in a narrower load range. The possibility to interface a single Li-Ion battery cell to a 48V bus was also demonstrated, even if in this case power losses appreciably increase.

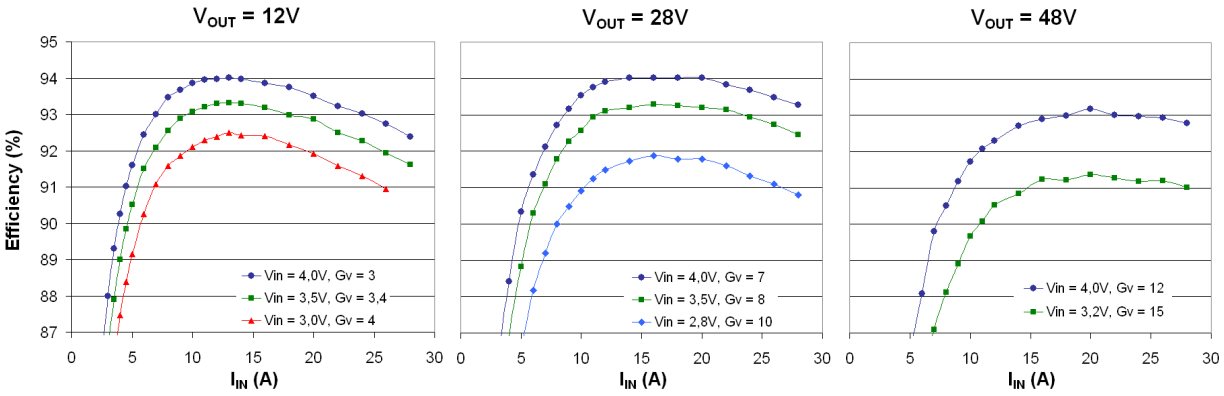


Figure 9. Overall efficiency Vs. input current for a 50 kHz converter operating in BDR-boost mode with different bus levels and input voltages (controller absorption is included).

The trick to obtain such high voltage ratio by a transformerless converter lies in an extreme care in reducing power losses [9]. In fact, the duty-cycle D required to achieve a specified voltage ratio G_V is:

$$D = 1 - \frac{1}{G_V} + \frac{\Delta}{G_V} \quad (1)$$

where the loss-dependent term Δ can be approximately estimated by the first-order expression:

$$\Delta = \frac{P_{LOSS} / I_{BAT}^2}{R_L / G_V^2 + P_{LOSS} / I_{BAT}^2} \quad (2)$$

Among the tests carried out on the demonstrator board, here we report the results obtained during the transitions between BCR to BDR modes. Step transients were forced by applying large variations of load absorption, obtaining the typical trend reported in Fig. 10. Before the transition, the power available from SA was enough to supply the bus and to charge battery with a constant current of about 5 A (green curve).

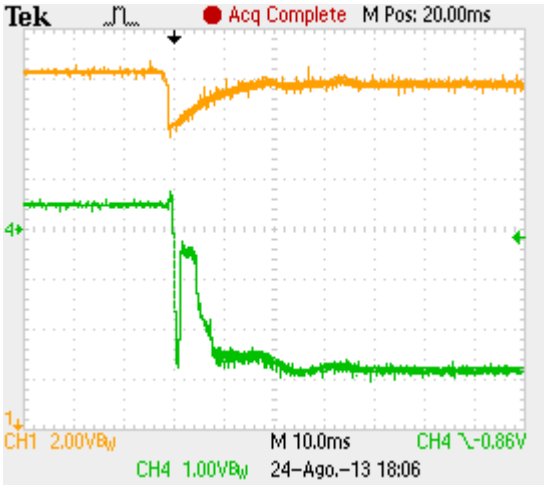


Figure 10. Typical transition from BCR to BDR operating mode, corresponding to a pulsed load transition during battery charge (from 5A in BCR mode to 28A in BDR mode).

After the load step, the power from solar array is no more sufficient to supply the bus, and the converter changes its operating mode from BCR to BDR, starting to regulate the bus with a battery discharge current of 28A. The bus has a transient voltage drop of about 2 V and reaches the new stationary condition within 30ms.

3. CONCLUSIONS

The power system of a small lunar rover developed for the AMALIA lunar mission has been described. In particular, an unusual approach for the battery system is reported: it is based on a single-cell battery system and a non-insulated high-voltage-diversity bidirectional converter as a BCDR. The reported experimental results demonstrated the feasibility of this approach up to 3/48V (16×) voltage gain at 90W. Some design issues related to the PCDU design have been examined.

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